Auteur: Thijs Van Schel
Studierichting: Informatica
Oefeningenreeks 7B nr. 3.1

$r = e^{u+v+w}$ met $u = yz$, $v = xz$ en $w = xy$

Partiële afgeleiden:

$$\begin{cases} \dfrac{\partial r}{\partial u} = \dfrac{\partial r}{\partial v} = \dfrac{\partial r}{\partial w} = e^{u+v+w} \\ \dfrac{\partial u}{\partial x} = 0, \dfrac{\partial u}{\partial y} = z, \dfrac{\partial u}{\partial z} = y \\ \dfrac{\partial v}{\partial x} = z, \dfrac{\partial v}{\partial y} = 0, \dfrac{\partial v}{\partial z} = x \\ \dfrac{\partial w}{\partial x} = y, \dfrac{\partial w}{\partial y} = x, \dfrac{\partial w}{\partial z} = 0 \end{cases}$$

$\Rightarrow$ uitwerken

$$\begin{cases} \dfrac{\partial r}{\partial x} = e^{u+v+w}(0) + e^{u+v+w}(z) + e^{u+v+w}(y) = e^{yz+xz+xy}(y+z) \\ \dfrac{\partial r}{\partial y} = e^{u+v+w}(z) + e^{u+v+w}(0) + e^{u+v+w}(x) = e^{yz+xz+xy}(x+z) \\ \dfrac{\partial r}{\partial z} = e^{u+v+w}(y) + e^{u+v+w}(x) + e^{u+v+w}(0) = e^{yz+xz+xy}(x+y) \end{cases}$$

References